

Stage 7 — How robust is the location law?

Where the Shortcut Sits: progress report 7

In one sentence. The location law does not depend on how masking is implemented — black fill, a blurred background, cropping to the lesion box and cropping the masked image all give a positive crossover in all four real-artifact cohorts (16 of 16, Holm) — it is graded rather than a threshold effect (the mask gain falls monotonically with the real overlap r in all four cohorts, and for hair turns into harm above $r \approx 0.401$), it reappears on an external, patient-level test set at natural prevalence (ISIC 2019 → ISIC 2020: masking leaves the overall AUROC nearly unchanged but lowers it on the shortcut-conflicting pairs, by -0.037 [$-0.061, -0.012$] under a calibrated duplicate rule), and the two-parameter theory of Stage 5, tested prospectively on 84 cells that did not exist when the test was registered, met its registered criterion for quantitative prediction.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma, malignant nodule, tumour with a solid component, erosion.
$A \in \{0, 1\}$	artifact present: a real artifact: hair, sonographer caliper, capsule debris.
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$C_v = [v - \text{ERM}]_{\text{Trap B}} - [v - \text{ERM}]_{\text{Trap A}}$ of reversed-test AUROC, for a masking implementation v (mean-colour fill in Stages 1–6).
CI	95% crossed seed×image bootstrap interval (Stage 1), 10,000 replicates; one Poisson weight per test image shared by every term in which the image appears.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
r	share of an artifact’s pixels that lie inside the ROI (Stage 3).
g_b	mask gain in overlap bin b : reversed AUROC of mask minus ERM.
Hard pairs	positive–negative pairs whose artifact status conflicts with the training association (Stage 5).

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1 Where this stage starts

Stages 1–6 established the location law — ROI masking removes an out-of-ROI shortcut and keeps, or amplifies, an in-ROI one — in controlled sweeps (Stage 2), with real artifacts in four cohorts (Stage 3), against the alternative that different images carry the artifact in the two traps (Stage 4), on natural test sets and across encoders, with a theory that closely matches held-out results (Stage 5), and they built remedies around it (Stage 6). Four objections remained open, each of which a reviewer can raise against the central claim:

1. *“Mean-colour fill is one implementation.”* Clinical pipelines crop to the lesion or use black backgrounds; the law might be a property of the particular fill used in Stages 1–6.
2. *“Trap A ($r \geq 0.5$) and Trap B ($r < 0.1$) are two hand-picked extremes.”* The crossover might be a dichotomy produced by the thresholds rather than a graded dependence on where the artifact lies.
3. *“Resampled traps are artificial and the data are internal.”* The natural test sets of Stage 5 were held-out parts of the training datasets; the ISIC 2020 validation of Stage 5 still used resampled traps.
4. *“The theory was checked only on results that already existed.”* Stage 5’s test of the theory was retrospective.

This stage answers each with one pre-registered analysis (R8–R11). The first external result raised two doubts of its own — an aggressive duplicate rule and mixed hair groups — which a second registration examined (R12, R13).

2 Questions

- Q7a** Does the crossover hold for other masking implementations, and do implementations that remove out-of-ROI content only partly give a weaker location effect?
- Q7b** Does the mask gain fall continuously as more of a real artifact lies inside the ROI?
- Q7c** Trained on one dataset and tested on another collected from other patients, at natural prevalence and without resampling, does masking harm the cases that conflict with the in-lesion hair association?
- Q7d** Does the theory predict results it has never seen, by the rule registered before they existed?
- Q7e** Does the external result survive a principled, label-free duplicate rule, and does it hold for the cleanest hair groups?

Design decisions *One registration for R8–R11* (Table 1), committed before any feature, cell count or model of them existed, and a second one for R12–R13 (4af106a, 2026-09-28 13:37), committed after R10 was known and before any analysis of R12–R13; R10 stays reported as registered, and R12–R13 are sensitivity analyses of it. Every hypothesis is reported whichever way it went. *Identical environments*: every trap of R8 and R9 uses the cohorts, artifact definitions, matching, seeds, folds, heads and thresholds of Stage 3, so the new arms are paired with ERM on the same images. *A reproducibility gate*: each R8 run re-fits ERM and mean-colour masking and had to reproduce the stored Stage 3 runs within 0.03 AUROC before any new result was read; the largest difference was 0.000. *Counts before models*: the dose-response bins, their count gate and the merge rule were applied and committed (ac90660, 2026-09-28 12:48) before any dose-response model was fitted (results committed d2a854b, 2026-09-28 13:13). *The crossed bootstrap from the start* (Stage 1), so no interval of this stage needs a second estimator check. Rejected: a new encoder for these analyses (it would confound the new question with a change of representation), and choosing the overlap bins after seeing the data.

3 Methods

R8 — masking implementations Five views of every image, computed from the cached 518-px image and its ROI (Figure 1): *mean-colour fill* (the reference of Stages 1–6); *black fill*; *blurred background* (pixels outside the ROI replaced by a Gaussian-blurred copy, $\sigma = 16$ px; the ROI stays sharp — a partial removal: coarse context survives, thin structures fade); *box crop* (the ROI bounding box enlarged by 10% on each side, padded to a square and resized — out-of-ROI pixels near the lesion survive); *box crop of the masked image* (no out-of-ROI pixels; the lesion is enlarged to fill the frame). A plain ERM head is fitted on each view. The primary family is the crossover C_v of the four new views in the four cohorts (16 tests, Holm). H8a: $C_v > 0$ for the two full-removal views in all four cohorts. H8b: the two partial-removal views also give $C_v > 0$, but a smaller one than mean fill ($C_{\text{mask}} - C_v > 0$). 0 images had an empty ROI.

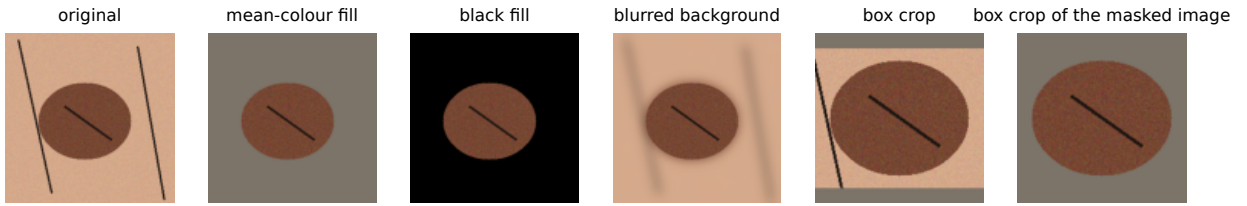


Figure 1: The masking implementations of R8, illustrated on a synthetic image (a lesion with a hair-like line inside it and two outside). No patient image is shown.

R9 — dose-response over the real overlap The artifact-bearing images of each cohort are binned by r : $[0, 0.1)$, $[0.1, 0.3)$, $[0.3, 0.5)$, $[0.5, 0.75)$, $[0.75, 1]$. Each bin is a trap with the cohort’s shared artifact-free group, built exactly like Trap A and Trap B. A bin with fewer than 25 images in any matched cell or fewer than 40 reversed-test positives is merged towards the middle of the r range by a fixed rule. The primary endpoint is the least-squares slope of g_b on the bin’s mean r , written as a linear combination of AUROCs so that the crossed bootstrap gives the artifact-free images, which appear in every bin, one shared weight. H9: slope < 0 in every cohort (Holm).

R10 — ISIC 2019 → ISIC 2020 Train and validate on all 25,179 quality-controlled ISIC 2019 images (4,509 melanomas) at their natural distribution; 5 seeds of a group-safe 80/20 train/validation split. Test on every ISIC 2020 image (Rotemberg et al., 2021) with a lesion mask (32,997) after removing near-duplicates of any ISIC 2019 image (perceptual hash, Hamming distance ≤ 8 — the leakage rule of Stage 1): 25,340 images of 2,044 patients, 483 melanomas. Lesion and hair masks are those of the Stage 5 external validation (models frozen before any ISIC 2020 image was scored). In-lesion hair (A): hair on the lesion with $r \geq 0.5$. Arms as in the natural test sets of Stage 5 plus the remedies of Stage 6. Operating points OP1–OP5 use thresholds from each seed’s ISIC 2019 validation predictions only. H10a: hard-pair AUROC mask – ERM < 0 ; H10b: at OP5 masking lowers sensitivity among melanomas without in-lesion hair; H10c: group balancing beats masking on hard pairs.

R12 — a calibrated duplicate rule All 32,997 ISIC 2020 images with a lesion mask are scored once with the R10 models, so that every duplicate rule is a subset of one set of predictions; on the R10 test images the predictions must equal R10’s (largest difference 2.2×10^{-7} over 1,393,700 rows). The rule is calibrated without labels and committed before any model was fitted (a02db23, 2026-09-28 14:01): the false-match rate at a hash distance t is the share of ISIC 2020 images that lie within t of an ISIC 2020 image of a *different patient* — pairs that are almost never copies — and likewise for the cosine similarity of DINOv2 features. The thresholds are the loosest ones with a false-match rate $\leq 1\%$: hash distance ≤ 1 or similarity ≥ 0.971 . H12: hard-pair AUROC mask

– ERM < 0 on the calibrated test set. Four rules are reported side by side: exact copies, distance ≤ 2 , calibrated, and distance ≤ 8 (R10).

R13 — cleaner hair groups On the calibrated test set: hair-free (at most τ predicted hair pixels), clear in-lesion hair (at least 3τ and $r \geq 0.5$), clear hair beside the lesion only (at least 3τ and $r < 0.1$), and ambiguous images, which are left out of these pairs. H13: AUROC mask – ERM < 0 for hair-free melanomas against benign lesions with clear in-lesion hair — the purest test of a retained in-ROI shortcut. Pairs involving hair beside the lesion are descriptive.

R11 — prospective theory test The procedure of Stage 5, unchanged: for every cell (cohort $\times$ trap or bin $\times$ arm) the two parameters (S, A) are fitted to the clean and correlated AUROC, and the reversed AUROC — and every crossover, gain and slope built from it — is predicted. The cells are the 84 R8 and R9 cells, none of which existed when the rule was registered. Decision rule as in Stage 5: *quantitatively predictive* only if every R8 crossover sign agrees, both correlations are ≥ 0.7 and the error beats “reversed = clean”.

4 Experiments

Table 1: Experiments of this stage. docs/PREREGISTRATION_ROUND4.md was committed before any feature, count or model of R8–R11 existed, docs/PREREGISTRATION_ROUND5.md before any analysis of R12–R13. Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	First commit (local time)	Support criterion
R8 masking implementations (H8a, H8b)	registered before results (round 4)	d34b8b8, 2026-09-28 11:28	H8a: $C_v > 0$ for black fill and box crop of the masked image in all four cohorts (Holm over 16)
R9 dose-response over the real overlap (H9)	registered before results (round 4); bin counts committed before fitting	d34b8b8, 2026-09-28 11:28	slope of the mask gain on $r < 0$ in every cohort that passes the count gate (Holm)
R10 ISIC 2019 $\rightarrow$ ISIC 2020 natural test (H10a–c)	registered before results (round 4)	d34b8b8, 2026-09-28 11:28	hard-pair AUROC mask – ERM < 0 ; OP5 sensitivity loss; balancing $>$ mask on hard pairs
R11 prospective theory test (T1'–T3')	registered before results (round 4)	d34b8b8, 2026-09-28 11:28	all crossover signs, $r \geq 0.7$, MAE below “reversed = clean”
R12 calibrated duplicate rule (H12)	registered before results (round 5); sensitivity analysis of R10, registered after R10; label-free calibration committed before fitting	4af106a, 2026-09-28 13:37	hard-pair mask – ERM CI < 0 on the calibrated test set (“robust”)
R13 cleaner hair groups (H13)	registered before results (round 5); secondary	4af106a, 2026-09-28 13:37	hair-free melanomas vs benign with clear in-lesion hair: CI < 0 ; gate $\geq 30 / \geq 100$

5 Results

5.1 R8: the law does not depend on how masking is implemented

Table 2 and Figure 2. Every implementation gives a positive crossover in every cohort: 16 of 16 tests are supported after Holm adjustment. For the full-removal views (H8a) this is 8 of 8, so by the registered rule the law is **implementation-independent**. The re-fitted mean-colour crossovers reproduce Stage 3 (dermoscopy $+0.147$ [$+0.111, +0.185$], thyroid $+0.239$ [$+0.196, +0.280$], capsule $+0.377$ [$+0.328, +0.426$], ovary $+0.170$ [$+0.088, +0.250$]).

Table 2: Location crossover C_v for every masking implementation (DINOv2 ViT-B/14, crossed 95% CIs). All 16 non-reference crossovers are positive with Holm-adjusted $p \leq 0.010$ (family of 16).

Cohort	mean-colour fill (reference)	black fill	blurred background	box crop	box crop of the masked image
Dermoscopy hair (ISIC 2019)	+0.147 [+0.111, +0.185]	+0.170 [+0.133, +0.207]	+0.154 [+0.117, +0.191]	+0.091 [+0.064, +0.117]	+0.154 [+0.121, +0.187]
Thyroid calipers	+0.239 [+0.196, +0.280]	+0.223 [+0.181, +0.265]	+0.218 [+0.177, +0.258]	+0.246 [+0.206, +0.285]	+0.250 [+0.210, +0.290]
Capsule debris	+0.377 [+0.328, +0.426]	+0.367 [+0.318, +0.416]	+0.278 [+0.221, +0.330]	+0.317 [+0.267, +0.367]	+0.358 [+0.305, +0.409]
Ovary calipers	+0.170 [+0.088, +0.250]	+0.153 [+0.063, +0.242]	+0.112 [+0.027, +0.195]	+0.145 [+0.083, +0.207]	+0.160 [+0.080, +0.240]

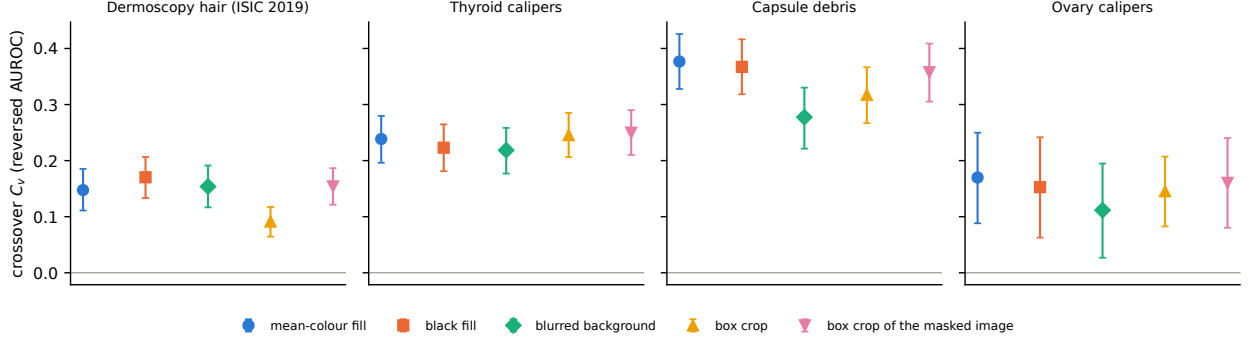


Figure 2: Crossover C_v per masking implementation and cohort (crossed 95% CIs). The reference (mean-colour fill) is the leftmost point of each panel.

Partial removal (H8b) Both partial-removal views also give positive crossovers in every cohort (8 of 8), but they are weaker than mean fill only where the surviving content still carries the artifact (Table 3). The blurred background weakens the effect for capsule debris (+0.099 [+0.067, +0.132]) and ovarian calipers (+0.058 [+0.004, +0.113]), not for hair (−0.006 [−0.042, +0.024]): a 16-px blur erases thin hair but leaves large debris recognisable. The box crop weakens it for hair (+0.056 [+0.012, +0.099]) and debris (+0.059 [+0.032, +0.087]), whose out-of-ROI instances often lie next to the lesion and survive the crop, but not for thyroid or ovarian calipers (−0.007 [−0.049, +0.032], +0.024 [−0.045, +0.096]). The registered expectation “partial removal is weaker” therefore holds in 4 of 8 contrasts. One result was not expected: for hair, black fill gives a *larger* crossover than mean fill (−0.023 [−0.040, −0.007] as $C_{\text{mask}} - C_v$).

Table 3: Secondary contrast $C_{\text{mask}} - C_v$: how much smaller the crossover of an implementation is than that of mean-colour fill (positive = weaker location effect than the reference).

Cohort	black fill	blurred background	box crop	box crop of the masked image
Dermoscopy hair (ISIC 2019)	−0.023 [−0.040, −0.007]	−0.006 [−0.042, +0.024]	+0.056 [+0.012, +0.099]	−0.007 [−0.033, +0.019]
Thyroid calipers	+0.016 [−0.004, +0.036]	+0.020 [−0.008, +0.049]	−0.007 [−0.049, +0.032]	−0.011 [−0.037, +0.014]
Capsule debris	+0.010 [−0.001, +0.021]	+0.099 [+0.067, +0.132]	+0.059 [+0.032, +0.087]	+0.019 [+0.001, +0.037]
Ovary calipers	+0.017 [−0.034, +0.079]	+0.058 [+0.004, +0.113]	+0.024 [−0.045, +0.096]	+0.010 [−0.046, +0.068]

What each implementation does in each trap Figure 3 and Table 4. The pattern of Stage 3 recurs for every implementation. For hair, every full-removal view harms inside the lesion (Trap A: mean fill −0.049 [−0.083, −0.017], black −0.061 [−0.097, −0.028]) and helps beside it (Trap B: mean fill +0.098 [+0.060, +0.136]). The views that keep some context harm less in Trap A (blur −0.023 [−0.051, +0.005], box crop −0.017 [−0.038, +0.003]) and gain less in Trap B (box crop +0.074 [+0.049, +0.098]). For calipers and debris every view still helps a little in Trap A — it removes clutter and overlays elsewhere on the screen (Stage 3) — and much more in Trap B.

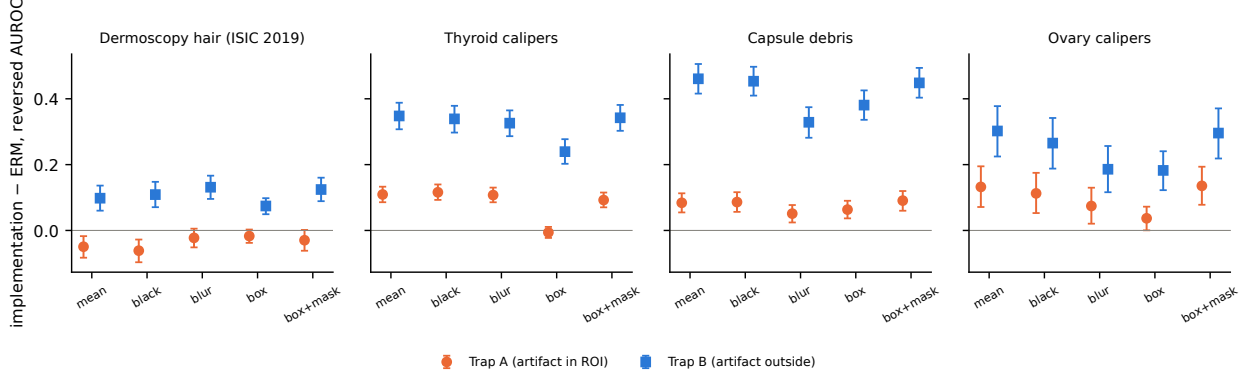


Figure 3: Each implementation minus ERM on the reversed test, in Trap A (artifact inside the ROI, orange circles) and Trap B (outside, blue squares); crossed 95% CIs.

Table 4: Each implementation minus ERM, per trap and test environment (AUROC, crossed 95% CIs; descriptive).

Cohort	Implementation	Trap A reversed	Trap B reversed	Trap A clean	Trap B clean
Dermoscopy hair (ISIC 2019)	mean-colour fill	-0.049 [-0.083, -0.017]	+0.098 [+0.060, +0.136]	-0.023 [-0.045, -0.001]	-0.014 [-0.041, +0.014]
	black fill	-0.061 [-0.097, -0.028]	+0.109 [+0.071, +0.148]	-0.028 [-0.052, -0.005]	-0.006 [-0.034, +0.022]
	blurred background	-0.023 [-0.051, +0.005]	+0.131 [+0.096, +0.166]	-0.004 [-0.022, +0.014]	+0.020 [-0.006, +0.047]
	box crop	-0.017 [-0.038, +0.003]	+0.074 [+0.049, +0.098]	-0.005 [-0.019, +0.009]	+0.010 [-0.007, +0.026]
	box crop of the masked image	-0.030 [-0.062, +0.002]	+0.124 [+0.089, +0.160]	-0.010 [-0.030, +0.011]	+0.011 [-0.014, +0.035]
Thyroid calipers	mean-colour fill	+0.109 [+0.086, +0.133]	+0.348 [+0.307, +0.388]	+0.081 [+0.063, +0.100]	+0.166 [+0.118, +0.213]
	black fill	+0.116 [+0.093, +0.140]	+0.339 [+0.297, +0.379]	+0.067 [+0.044, +0.089]	+0.171 [+0.125, +0.218]
	blurred background	+0.107 [+0.086, +0.130]	+0.326 [+0.286, +0.365]	+0.065 [+0.044, +0.085]	+0.154 [+0.109, +0.199]
	box crop	-0.006 [-0.023, +0.011]	+0.239 [+0.203, +0.277]	+0.015 [-0.003, +0.032]	+0.087 [+0.040, +0.135]
	box crop of the masked image	+0.092 [+0.070, +0.115]	+0.342 [+0.303, +0.381]	+0.083 [+0.063, +0.101]	+0.174 [+0.129, +0.219]
Capsule debris	mean-colour fill	+0.084 [+0.055, +0.113]	+0.461 [+0.416, +0.506]	+0.064 [+0.044, +0.086]	+0.217 [+0.186, +0.251]
	black fill	+0.086 [+0.056, +0.116]	+0.453 [+0.410, +0.497]	+0.063 [+0.044, +0.084]	+0.215 [+0.183, +0.248]
	blurred background	+0.051 [+0.024, +0.077]	+0.329 [+0.282, +0.374]	+0.047 [+0.027, +0.068]	+0.156 [+0.129, +0.184]
	box crop	+0.063 [+0.037, +0.090]	+0.381 [+0.336, +0.425]	+0.046 [+0.024, +0.068]	+0.174 [+0.145, +0.206]
	box crop of the masked image	+0.091 [+0.060, +0.120]	+0.448 [+0.403, +0.494]	+0.056 [+0.033, +0.079]	+0.205 [+0.173, +0.238]
Ovary calipers	mean-colour fill	+0.132 [+0.071, +0.195]	+0.302 [+0.225, +0.378]	+0.055 [+0.014, +0.095]	+0.097 [+0.032, +0.159]
	black fill	+0.113 [+0.053, +0.175]	+0.265 [+0.188, +0.342]	+0.028 [-0.013, +0.069]	+0.093 [+0.037, +0.149]
	blurred background	+0.074 [+0.021, +0.130]	+0.186 [+0.116, +0.257]	+0.050 [+0.012, +0.086]	+0.065 [+0.014, +0.117]
	box crop	+0.037 [+0.000, +0.072]	+0.182 [+0.122, +0.241]	+0.041 [+0.014, +0.068]	+0.078 [+0.038, +0.117]
	box crop of the masked image	+0.135 [+0.078, +0.194]	+0.296 [+0.219, +0.371]	+0.050 [+0.007, +0.092]	+0.107 [+0.046, +0.167]

5.2 R9: the location effect is graded

Table 5, Table 6 and Figure 4. In every cohort the mask gain falls as more of the artifact lies inside the ROI: slope $-0.235 [-0.276, -0.194]$ (hair), $-0.250 [-0.303, -0.197]$ (thyroid), $-0.489 [-0.539, -0.438]$ (capsule) and $-0.205 [-0.300, -0.109]$ (ovary); 4 of 4 supported after Holm adjustment, with a perfectly monotone ordering of the bins in every cohort. The two traps of Stages 3–6 are the ends of a continuous dependence, not an artefact of the thresholds. For hair the gain passes through zero near $r = 0.401$ and becomes a harm when almost all of the hair lies on the lesion ($-0.086 [-0.119, -0.056]$ for $r \geq 0.75$, against $+0.124 [+0.085, +0.162]$ for $r < 0.1$). For calipers and debris the gain shrinks several-fold (capsule $+0.465 [+0.422, +0.508] \rightarrow +0.032 [-0.004, +0.069]$; thyroid $+0.336 [+0.295, +0.377] \rightarrow +0.116 [+0.091, +0.141]$) but stays non-negative, for the reason given above: masking also removes on-screen clutter. The ovary cohort had too few images with intermediate overlap and was analysed with three bins after the registered merge rule (3 bins).

Table 5: Dose-response: least-squares slope of $g_b = \text{rev}(\text{mask}) - \text{rev}(\text{ERM})$ on the bin mean overlap r (crossed 95% CIs; the artifact-free images shared by all bins carry one bootstrap weight).

Cohort	Bins	Slope of the mask gain on r	Holm p	Spearman	Zero crossing r	H9
Dermoscopy hair (ISIC 2019)	5	$-0.235 [-0.276, -0.194]$	< 0.001	-1.00	0.401	supported
Thyroid calipers	5	$-0.250 [-0.303, -0.197]$	< 0.001	-1.00	none	supported
Capsule debris	5	$-0.489 [-0.539, -0.438]$	< 0.001	-1.00	none	supported
Ovary calipers	3	$-0.205 [-0.300, -0.109]$	< 0.001	-1.00	none	supported

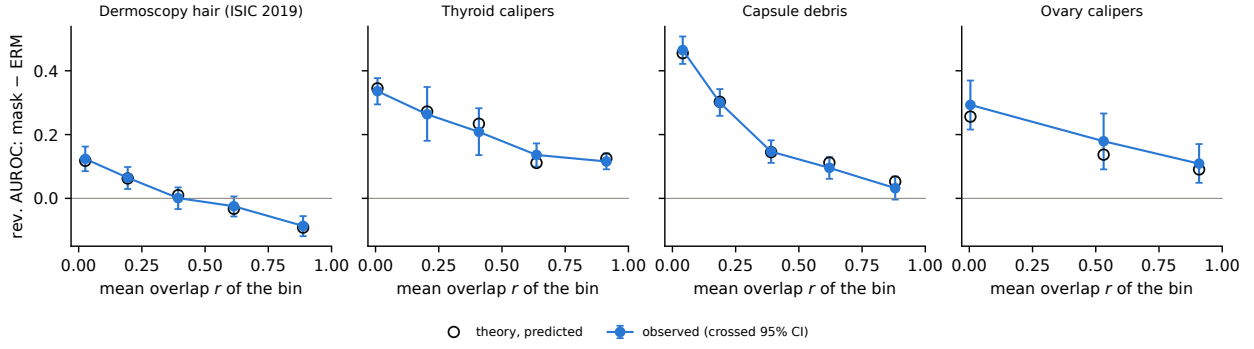


Figure 4: Mask gain on the reversed test against the mean overlap r of each bin (filled: observed, crossed 95% CIs; open: the theory’s prediction from each bin’s clean and correlated AUROC only).

Table 6: Bins after the registered count gate (≥ 25 per matched cell, ≥ 40 reversed-test positives over the folds of seed 42); merged bins are named by their parts. Cell counts are those of the matched pool.

Cohort	Bin	r range	Mean r	$A=1: Y=0 / Y=1$	$A=0: Y=0 / Y=1$	Rev. positives	Mask gain g_b
Dermoscopy hair (ISIC 2019)	b1	0.00–0.10	0.026	4412 / 969	706 / 157	172	+0.124 [+0.085, +0.162]
	b2	0.10–0.30	0.194	3749 / 797	706 / 157	172	+0.065 [+0.029, +0.098]
	b3	0.30–0.50	0.393	2458 / 762	706 / 157	172	+0.001 [−0.034, +0.035]
	b4	0.50–0.75	0.614	1579 / 612	706 / 157	172	−0.025 [−0.057, +0.006]
	b5	0.75–1.00	0.888	1074 / 329	706 / 157	172	−0.086 [−0.119, −0.056]
Thyroid calipers	b1	0.00–0.10	0.008	227 / 99	1052 / 749	829	+0.336 [+0.295, +0.377]
	b2	0.10–0.30	0.204	75 / 30	1052 / 749	300	+0.264 [+0.180, +0.349]
	b3	0.30–0.50	0.409	70 / 40	1052 / 749	400	+0.208 [+0.136, +0.283]
	b4	0.50–0.75	0.637	229 / 79	1052 / 749	790	+0.136 [+0.099, +0.172]
	b5	0.75–1.00	0.913	629 / 213	1052 / 749	829	+0.116 [+0.091, +0.141]
Capsule debris	b1	0.00–0.10	0.041	231 / 1009	164 / 343	378	+0.465 [+0.422, +0.508]
	b2	0.10–0.30	0.188	391 / 769	164 / 343	378	+0.300 [+0.259, +0.343]
	b3	0.30–0.50	0.390	279 / 311	164 / 343	378	+0.147 [+0.111, +0.182]
	b4	0.50–0.75	0.621	237 / 227	164 / 343	378	+0.096 [+0.061, +0.130]
	b5	0.75–1.00	0.881	167 / 185	164 / 343	378	+0.032 [−0.004, +0.069]
Ovary calipers	b1	0.00–0.10	0.004	91 / 103	170 / 178	196	+0.293 [+0.216, +0.369]
	b2b3b4	0.10–0.75	0.531	59 / 55	170 / 178	196	+0.179 [+0.091, +0.266]
	b5	0.75–1.00	0.909	298 / 244	170 / 178	196	+0.109 [+0.049, +0.170]

5.3 R10: an external test at natural prevalence

The data After near-duplicate removal the test set has 25,340 images (483 melanomas, 2,044 patients). The removal was large: 7,657 images (23%) were within Hamming distance 8 of an ISIC 2019 image — 26 at distance 0, 112 at 1–2, 790 at 3–5 and 6,729 at 6–8 — and it removed melanomas less often (17%) than benign lesions (23%). Most removed images are therefore look-alikes rather than copies; R12 replaces the rule (Section 5.4). The association that masking can amplify is present in both datasets: in-lesion hair on 30.5% of ISIC 2019 melanomas and 19.6% of benign lesions, and on 16.1% and 7.6% in ISIC 2020.

AUROC Tables 7 and 8, Figure 5. On all ISIC 2020 pairs masking *improves* the AUROC (+0.020 [+0.003, +0.038]; ERM 0.791, mask 0.811) — masking is the best arm by overall AUROC. The improvement comes entirely from the shortcut-aligned pairs (+0.045 [+0.005, +0.087]); on the shortcut-conflicting pairs masking lowers the AUROC (−0.026 [−0.052, −0.000]; H10a **supported**, but only just: the upper bound is −0.0004; R12 shows that this margin was an effect of the duplicate rule). This is the Stage 5 signature — right for the wrong reason on aligned cases, wrong on conflicting ones — on patients the model never saw, without any resampling. An external validation that reported only the overall AUROC would recommend masking. Group balancing recovers the conflicting pairs (+0.059 [+0.035, +0.084] over mask; H10c **supported**) at a cost on aligned pairs (−0.094 [−0.138, −0.051]) and overall (−0.032 [−0.051, −0.014]); combining masking with group balancing gives the highest cross-group AUROC (0.776, against ERM 0.772 and mask 0.746).

Table 7: ISIC 2020 (external, natural prevalence), models trained on ISIC 2019: seed means. Hard pairs = melanomas without in-lesion hair vs benign lesions with it (they conflict with the training association).

Arm	AUROC	Cross-group AUROC	AUROC, in-lesion hair	Hard pairs	Easy pairs
ERM	0.791	0.772	0.779	0.772	0.798
mask	0.811	0.746	0.781	0.746	0.843
balanced	0.779	0.749	0.776	0.805	0.749
mask + balanced	0.792	0.776	0.777	0.786	0.783
U-MtE	0.799	0.752	0.783	0.752	0.827
U-MtE balanced	0.778	0.764	0.780	0.787	0.764
U-MtE protect	0.804	0.715	0.752	0.715	0.839
U-MtE protect balanced	0.790	0.753	0.747	0.753	0.789
DFR	0.765	0.730	0.746	0.782	0.731
mask + DFR	0.767	0.717	0.730	0.718	0.781
JTT	0.706	0.632	0.722	0.796	0.632

Table 8: ISIC 2020: AUROC contrasts with crossed 95% CIs (5 seeds; one Poisson weight per ISIC 2020 image).

Contrast	All pairs	Hard pairs	Easy pairs
mask – ERM	+0.020 [+0.003, +0.038]	−0.026 [−0.052, −0.000]	+0.045 [+0.005, +0.087]
balanced – mask	−0.032 [−0.051, −0.014]	+0.059 [+0.035, +0.084]	−0.094 [−0.138, −0.051]
U-MtE balanced – mask	−0.033 [−0.040, −0.026]	+0.041 [+0.030, +0.051]	−0.079 [−0.099, −0.059]
U-MtE – mask	−0.012 [−0.018, −0.007]	+0.005 [−0.002, +0.013]	−0.016 [−0.029, −0.003]

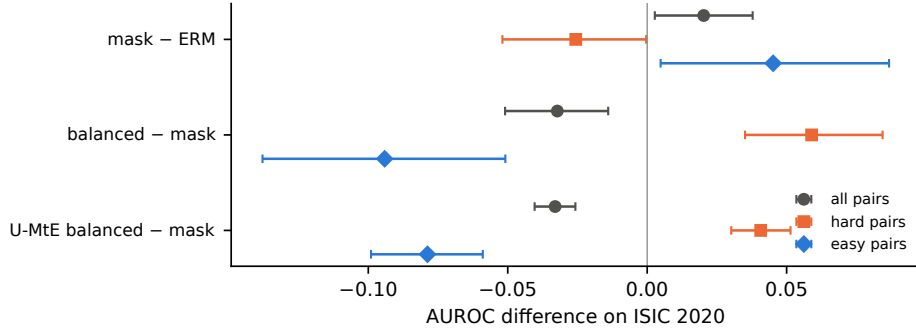


Figure 5: ISIC 2020: AUROC differences on all, hard (shortcut-conflicting) and easy (shortcut-aligned) pairs; crossed 95% CIs.

Operating points Table 9. The AUROC effect does *not* translate into a detectable sensitivity loss at a fixed threshold: at OP5 (validation sensitivity ≥ 0.90) the sensitivity among melanomas without in-lesion hair changes by $-0.007 [-0.060, +0.047]$ (H10b **not supported**); at OP1 masking even raises sensitivity without significance ($+0.082 [-0.014, +0.167]$), and at OP2–OP4 it raises specificity (OP2 $+0.029 [+0.017, +0.040]$). The clinical consequence shown on the thyroid split of Stage 5 is therefore not reproduced here.

Table 9: ISIC 2020 operating points, mask – ERM (crossed bootstrap with thresholds re-estimated on the weighted validation images in every replicate, 2,000 replicates).

Operating point (threshold from ISIC 2019 validation)	Sensitivity ERM → mask	Δ sensitivity	Δ specificity	Sensitivity, melanomas without in-lesion hair	Δ
OP1 max. balanced accuracy	0.434 → 0.516	+0.082 [−0.014, +0.167]	−0.003 [−0.046, +0.035]	0.432 → 0.519	+0.087 [−0.011, +0.168]
OP2 specificity ≥ 0.80	0.472 → 0.477	+0.005 [−0.033, +0.052]	+0.029 [+0.017, +0.040]	0.470 → 0.482	+0.012 [−0.031, +0.063]
OP3 specificity ≥ 0.90	0.309 → 0.324	+0.015 [−0.026, +0.062]	+0.016 [+0.009, +0.023]	0.309 → 0.332	+0.023 [−0.024, +0.072]
OP4 sensitivity ≥ 0.80	0.523 → 0.545	+0.022 [−0.033, +0.077]	+0.021 [+0.002, +0.039]	0.524 → 0.547	+0.022 [−0.031, +0.079]
OP5 sensitivity ≥ 0.90	0.702 → 0.699	−0.003 [−0.055, +0.052]	+0.028 [−0.011, +0.060]	0.700 → 0.693	−0.007 [−0.060, +0.047]

5.4 R12–R13: the external result, re-examined

Duplicates The calibration shows why the registered rule of R10 removed so much: 41% of ISIC 2020 images lie within hash distance 8 of an image of a *different patient* in ISIC 2020 itself — at that distance the hash matches similar-looking dermoscopy images, not copies. The calibrated rule removes 45 images and keeps 32,952 (575 melanomas). Table 10 and Figure 6.

Hard pairs (H12) With the calibrated rule the harm on shortcut-conflicting pairs is clear: mask – ERM $-0.037 [-0.061, -0.012]$ (H12 **supported**; decision label **robust**). It is the same under the exact-copy rule ($-0.037 [-0.061, -0.012]$) and at distance ≤ 2 ($-0.037 [-0.062, -0.013]$); the borderline R10 estimate ($-0.026 [-0.052, -0.000]$) was an effect of the aggressive rule, which removed a quarter of the test set. The shortcut-aligned pairs improve ($+0.048 [+0.009, +0.090]$), and the overall AUROC now barely moves ($+0.016 [-0.000, +0.033]$; lower bound -0.0003): an external validation reporting only the overall AUROC would see no problem. Group balancing again recovers the conflicting pairs ($+0.072 [+0.048, +0.097]$) at an overall cost ($-0.027 [-0.045, -0.010]$). At OP5 the sensitivity among melanomas without in-lesion hair still does not change detectably ($-0.022 [-0.076, +0.039]$).

Table 10: ISIC 2020 under four duplicate rules, from one set of predictions for all 32,997 images (crossed 95% CIs). The calibrated rule is label-free and was committed before any model of round 5 was fitted.

Duplicate rule	Removed	Test images	Melanomas	mask – ERM, all pairs	hard pairs	easy pairs	balanced – mask, hard pairs
exact copies only	26	32,971	578	+0.017 $[-0.000, +0.033]$	$-0.037 [-0.061, -0.012]$	+0.048 $[+0.009, +0.090]$	+0.072 $[+0.048, +0.096]$
hash distance ≤ 2	138	32,859	575	+0.016 $[-0.000, +0.032]$	$-0.037 [-0.062, -0.013]$	+0.048 $[+0.008, +0.090]$	+0.072 $[+0.048, +0.097]$
calibrated (R12 primary)	45	32,952	575	+0.016 $[-0.000, +0.033]$	$-0.037 [-0.061, -0.012]$	+0.048 $[+0.009, +0.090]$	+0.072 $[+0.048, +0.097]$
hash distance ≤ 8 (R10)	7,657	25,340	483	+0.020 $[+0.003, +0.038]$	$-0.026 [-0.052, -0.000]$	+0.045 $[+0.005, +0.087]$	+0.059 $[+0.035, +0.084]$

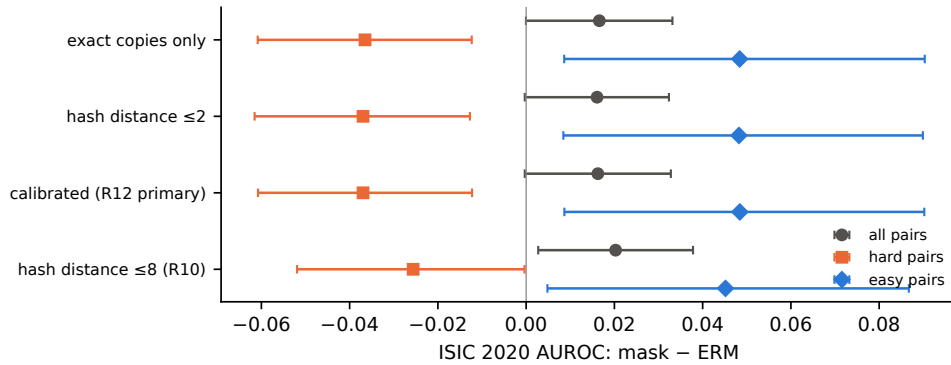


Figure 6: ISIC 2020, mask – ERM on all, hard and easy pairs under four duplicate rules (crossed 95% CIs).

Cleaner hair groups (H13) Table 11. The purest in-ROI comparison — 53 hair-free melanomas against 1,868 benign lesions with clear in-lesion hair — shows no detectable effect ($-0.005 [-0.057, +0.048]$; one-sided $p = 0.44$; H13 **not supported**); with only 53 hair-free melanomas its interval is wide. The descriptive pairs show where the hard-pair harm comes from. Melanomas with hair *beside* the lesion lose ground to hair-free benign lesions after masking ($-0.028 [-0.060, +0.004]$), and hair-free melanomas gain ground on benign lesions with hair beside them ($+0.059 [+0.010, +0.112]$): the model used hair beside the lesion as a melanoma cue, and masking removed it — the out-of-ROI half of the law, on natural data. Of the melanomas without in-lesion hair, 209 have clear hair beside the lesion. The hard-pair harm therefore combines both halves of the law — masking removes the melanomas’ hair beside the lesion and keeps the benign lesions’ hair on it — and is not evidence for amplification of in-lesion hair on its own.

Table 11: Cleaner hair groups on the calibrated ISIC 2020 test set (R13): hair-free ($\leq \tau$ predicted hair pixels), clear in-lesion hair ($\geq 3\tau$, $r \geq 0.5$) and clear hair beside the lesion only ($\geq 3\tau$, $r < 0.1$); ambiguous images excluded from these pairs.

Positives	Negatives	Contrast	AUROC difference [95% CI]	Status
hair-free melanomas	benign, clear in-lesion hair	mask – ERM	−0.005 [−0.057, +0.048]	H13
melanomas, clear in-lesion hair	hair-free benign	mask – ERM	+0.022 [−0.019, +0.064]	descriptive
melanomas, hair beside the lesion only	hair-free benign	mask – ERM	−0.028 [−0.060, +0.004]	descriptive
hair-free melanomas	benign, hair beside the lesion only	mask – ERM	+0.059 [+0.010, +0.112]	descriptive
hair-free melanomas	benign, clear in-lesion hair	balanced mask	+0.016 [−0.039, +0.075]	descriptive

5.5 R11: the theory predicts results it had not seen

Table 12 and Figure 7. Fitted only to each cell’s clean and correlated AUROC, the theory’s predictions of the 84 held-out reversed AUROCs have a mean absolute error of 0.011 (Pearson $r = 0.996$; 100% within 0.05), against 0.036 for the symmetric-reversal heuristic and 0.177 for “reversed = clean”. It gets the sign of all 16 R8 crossovers right (16 of 16; $r = 0.992$, MAE 0.011), the sign of 4 of 4 dose-response slopes (MAE 0.018) and of 18 of 18 bin gains. By the registered rule the verdict is **quantitatively predictive**. The difference from Stage 5’s verdict (a qualitative account) is one of scope, not of rule: Stage 5’s cell set included chest radiographs, where one near-zero crossover had the wrong sign; the cells here are frozen DINOv2 encoders with linear heads on four cohorts — the setting the theory assumes.

Table 12: Prospective theory test: fitted to each cell’s clean and correlated AUROC only; the reversed AUROC and every quantity derived from it are held out. None of these cells existed when the rule was registered.

Endpoint	Cells	Theory	“reversed = clean”	Symmetric reversal
T1’ reversed AUROC, all new cells	84	MAE 0.011, $r = 0.996$	MAE 0.177	MAE 0.036
T2’ R8 crossovers	16	signs 16/16, $r = 0.992$, MAE 0.011		
T3’ R9 slopes	4	signs 4/4, MAE 0.018		
T3’ R9 bin gains	18	signs 18/18, MAE 0.014		

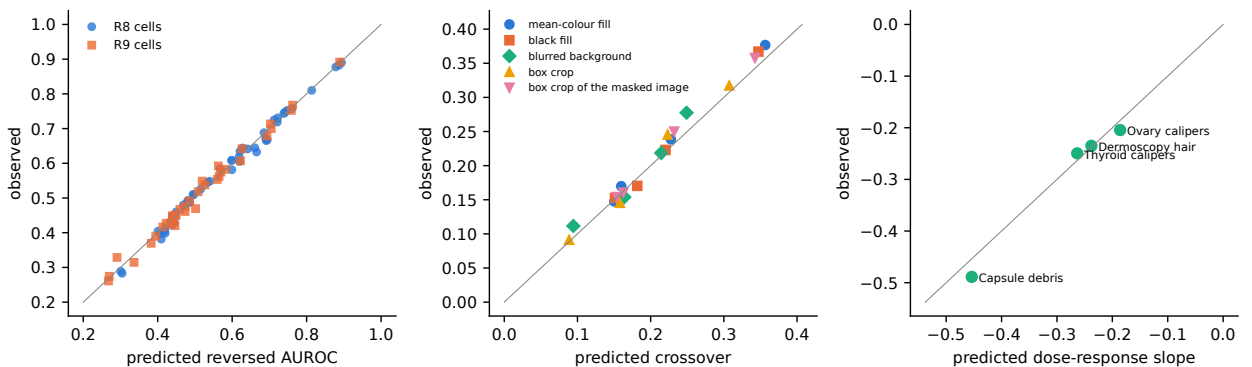


Figure 7: Prospective test of the theory. Left: predicted versus observed reversed AUROC for every R8 and R9 cell. Middle: R8 crossovers. Right: R9 dose-response slopes. Grey lines are the identity.

6 What failed or changed

- **H8b** holds in 4 of 8 contrasts: partial removal is weaker than mean fill for debris and ovarian calipers (blur) and for hair and debris (box crop), not otherwise; black fill gives a larger hair crossover than mean fill.
- **H10a** was supported only by a margin that disappears at the third decimal; under the calibrated duplicate rule (R12) the effect is clear. **H10b** is not supported under any rule: on ISIC 2020 masking does not measurably lower sensitivity at a fixed operating point.
- **The registered duplicate rule of R10** was too loose: it discarded 23% of ISIC 2020, mostly look-alikes; R12 replaced it with a label-free calibrated rule registered after R10 was known.
- **H13** is not supported: the cleanest in-ROI comparison has only 53 hair-free melanomas and shows no effect; the external hard-pair harm cannot be attributed to retained in-lesion hair alone.
- **The overall ISIC 2020 AUROC gain** from masking is borderline under the calibrated rule.
- **The ovary** dose-response has only three bins.
- **Scope:** R8–R10 use one encoder (DINOv2 ViT-B/14); R10 is dermoscopy only; the prospective theory test covers frozen-encoder linear heads only.
- No deviation from the registration. During the run only the execution was changed (parallel head fitting and bootstraps, one BLAS thread per process, feature extraction in a separate process, batch size 128 for the new views); data, seeds, models, estimands and bootstrap replicates are those registered, and the reproducibility gate was passed exactly.

7 Anticipated questions

Why test several masking implementations if the theory already says what masking does? Because the theory treats masking abstractly (it lowers the disease signal and removes the out-of-ROI artifact signal). Real implementations differ in what they leave behind — a black border, blurred context, nearby pixels — and a reviewer can reasonably suspect that one of them escapes the law. None does.

Why does blur work as well as mean fill for hair but not for debris? A 16-px Gaussian blur removes structures a few pixels wide — hair — almost completely, so for hair it is effectively full removal. Capsule debris and bubbles are large, low-frequency blobs that remain recognisable after blurring, so part of the out-of-ROI shortcut survives (+0.099 [+0.067, +0.132] weaker crossover).

Why is black fill stronger than mean fill for hair? Not established. Black fill removes slightly more disease context than mean fill inside Trap A (clean AUROC -0.028 [$-0.052, -0.005$] against -0.023 [$-0.045, -0.001$]) and gains slightly more in Trap B, which widens the crossover; why a black background does this for DINOv2 was not examined.

Why do the thyroid, capsule and ovary gains stay positive at $r \approx 0.9$? Masking removes more than the artifact: on-screen annotations, scanner overlays and clutter outside the ROI also carry label information in the training environment (Stage 3). The location law is about the part of the gain that the artifact’s position controls, and that part falls monotonically; for hair, where masking also removes disease context (Stage 2), the net gain becomes a harm.

Why is masking the best arm on ISIC 2020 by overall AUROC if it amplifies a shortcut? Because the shortcut agrees with the label in ISIC 2020 too: in-lesion hair is more common on melanomas. A model that relies more on in-lesion hair gains on the many aligned pairs and loses

on the fewer conflicting ones, and the aligned pairs dominate the overall AUROC. This is exactly why the location law matters for external validation: the harm is invisible in the headline metric.

Why is the AUROC harm not visible at the operating points? The hard-pair AUROC compares melanomas without hair with benign lesions *with* hair; a fixed threshold instead compares each melanoma with a threshold learned on ISIC 2019. Masking moved that threshold (specificity rises at OP2–OP4), and with 483 melanomas the sensitivity intervals are wide. On the thyroid split of Stage 5 the loss was large enough to show at every operating point; on ISIC 2020 it is not.

Why were so many ISIC 2020 images removed, and does it matter? The registered rule removed any ISIC 2020 image within hash distance 8 of any ISIC 2019 image, the threshold used to group near-duplicates for leakage-safe splits (Stage 1). Dermoscopy images share a global layout (a central dark lesion on skin), and R12 measured the consequence: 41% of ISIC 2020 images are that close to an image of another patient. Only 26 removed images were exact copies. With a label-free calibrated rule 45 images are removed, and the hard-pair harm is clearer, not weaker (-0.037 [$-0.061, -0.012$]). The loose rule cost power; it did not create the effect.

Is it legitimate to change the duplicate rule after seeing R10? Only as a registered sensitivity analysis, which is what R12 is: the new rule was fixed from label-free false-match rates and committed before any model was fitted, R10 is reported unchanged, and all four rules are shown. Every rule stricter than R10’s gives the same estimate.

Why does the cleanest in-ROI comparison show nothing? It rests on 53 hair-free melanomas, so its interval spans about ± 0.05 and cannot distinguish a small harm from none. The comparisons that involve hair beside the lesion are larger and behave as the out-of-ROI half of the law predicts. What the external data establish is therefore that masking changes the model’s use of hair according to where the hair is — not, on their own, that it amplifies in-lesion hair.

Does the “quantitatively predictive” verdict replace Stage 5’s “qualitative account”? No. Each verdict belongs to its registered cell set. The defensible statement is: across all cohorts (Stage 5) the theory is a qualitative account; for frozen-encoder linear heads on the four real-artifact cohorts it predicted 84 new cells prospectively within 0.011 AUROC and met the registered criterion.

8 Conclusion

Supported. The location crossover is independent of the masking implementation (16 of 16 tests; H8a 8 of 8); it is graded in the real overlap in all four cohorts (H9); on an external, patient-level test set at natural prevalence masking lowers the AUROC on shortcut-conflicting pairs (H10a; robust under a calibrated duplicate rule, H12) while the overall AUROC barely moves, and group balancing recovers those pairs (H10c); the theory met its registered criterion for quantitative prediction on 84 prospective cells. **Qualified.** Partial-removal views are weaker than mean fill in only 4 of 8 contrasts (H8b); the external harm is small and combines both halves of the law. **Not supported.** A sensitivity loss at a fixed operating point on ISIC 2020 (H10b); a detectable effect in the cleanest in-ROI hair comparison (H13).

9 Questions for the supervisor

1. Should the masking-implementation result (R8) and the dose-response (R9) enter the main text, as the direct answers to “is mean fill a strawman?” and “are the traps hand-picked?”?

2. Is the ISIC 2020 result — little overall change, conflicting-pair harm, no operating-point effect — best presented as an argument for subgroup reporting in external validation rather than as a clinical harm?
3. May the paper now say that the theory *predicts* for frozen-encoder linear probes, keeping “qualitative account” for the full cell set?
4. Which duplicate rule should the paper lead with: the registered rule of R10 (borderline) or the calibrated rule of R12 (robust), with the other reported alongside?

References

Veronica Rotemberg, Nicholas Kurtansky, Brigid Betz-Stablein, et al. A patient-centric dataset of images and metadata for identifying melanomas using clinical context. *Scientific Data*, 8:34, 2021.